

A *project specification*, or *spec*, lists requirements for what a program needs to do.

A clear specification is important in projects of all sizes so you have a solid idea of what to aim toward and whether your project is successful. Without a clear spec, you'll waste time and risk your project's failure.

A project specification indicates the problems that need solving and how users will interact with the program. It should specify what kinds of inputs your program will deal with, as well as the outputs the program needs to generate.

When learning to program, each exercise you attempt is a mini spec. Well-specified problems are easier to solve than poorly specified exercises. A good programmer looks for a spec when collaborating on an existing project and develops a full spec before committing to a new project.